

Predicting FOMC Policy Stance from Minutes

- Question: Can public FOMC minutes be converted into a measurable hawkish-dovish policy stance index?
- Data: Federal Reserve minutes from 2008-2026 and FRED Treasury yields.
- Method: sentence-level NLP classification plus document-level aggregation.

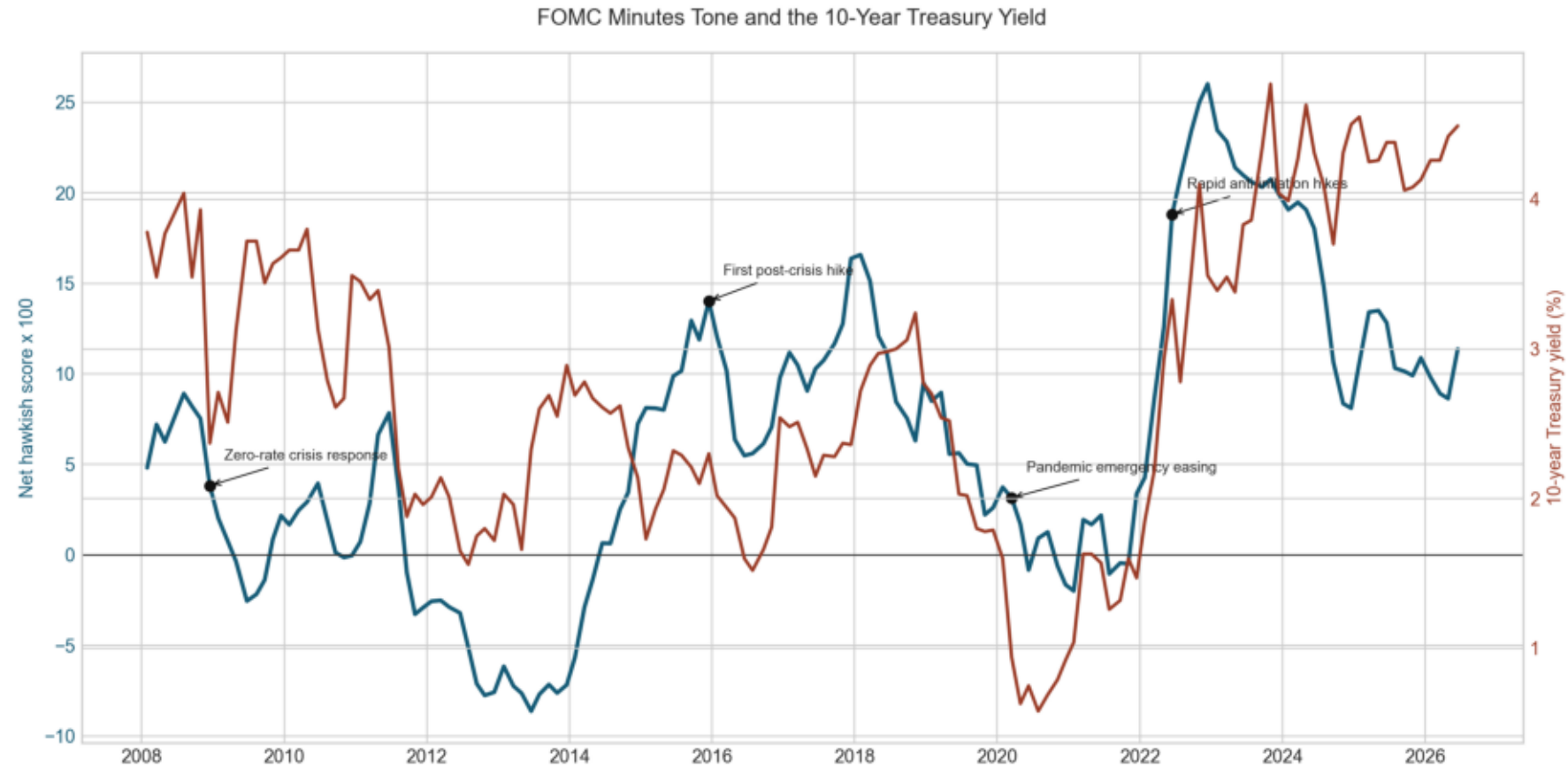
Pipeline

- Download official FOMC minutes and Treasury yield CSV files.
- Clean HTML text, split minutes into sentences, and classify each sentence.
- Aggregate net hawkish score and compare it with DGS10 and DGS2.

Sentence Classifier

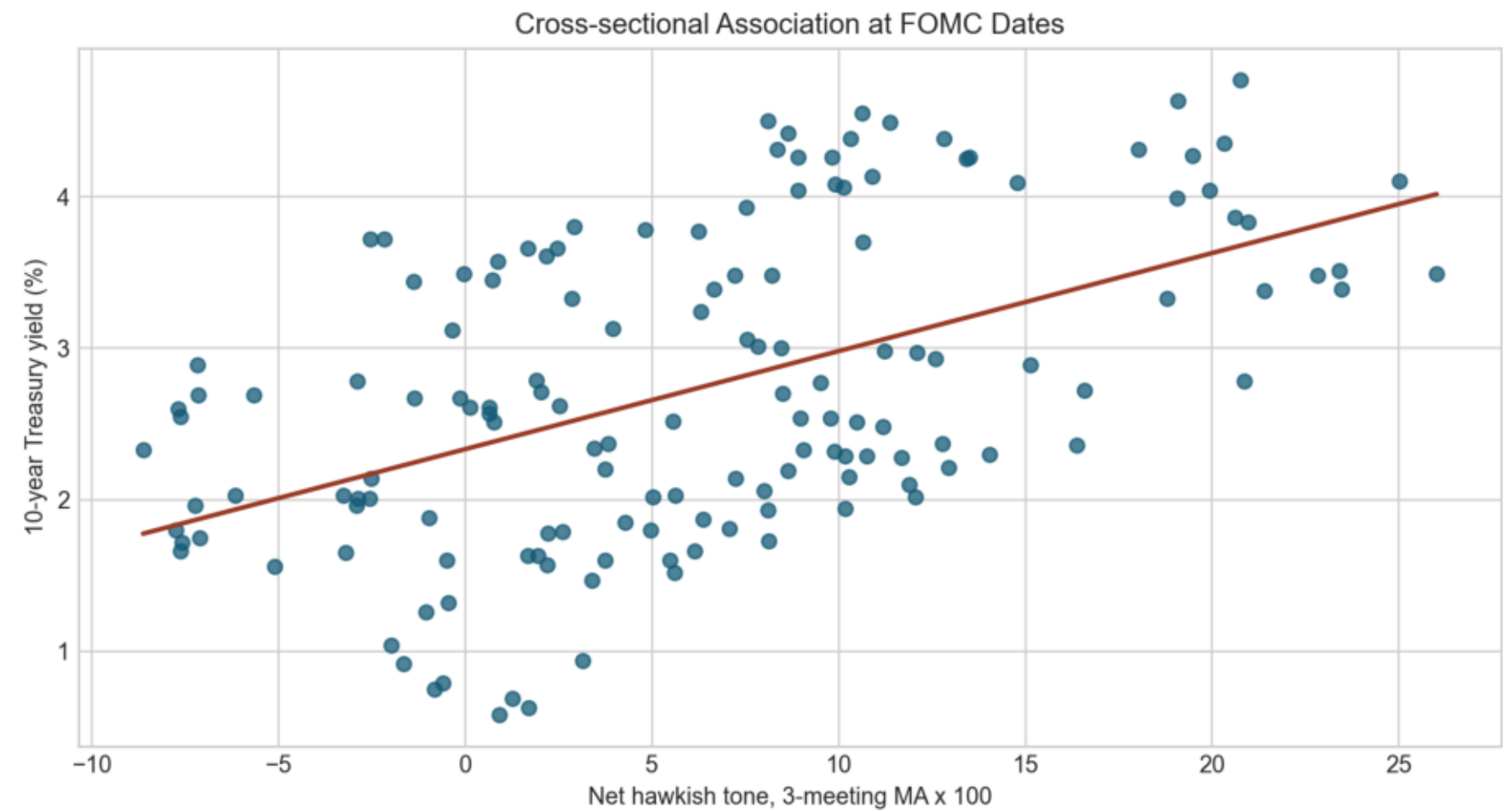
- Hawkish examples: inflation pressure, tightening, restrictive policy, rate increase.
- Dovish examples: accommodation, easing, downside risk, slack, asset purchases.
- Score = (hawkish sentences - dovish sentences) / total sentences.

Main Result



The tone index shifts are associated with crisis and tightening regimes, including 2008, 2020, and 2022.

Market Validation



The smoothed hawkish is positively related to the 10-year Treasury yield.

Quantitative Evidence

- Strongest validation: DGS2, tone lagged by 2 meetings.
- Pearson correlation $r = 0.806$; p-value = $2.36e-34$.
- The 2-year yield relationship is strongest, consistent with near-term policy sensitivity.

Conclusion

- FOMC minutes contain measurable policy-tone information.
- The dictionary model is transparent and reproducible.
- FinBERT fine-tuning is the natural next step for context, negation, and stronger sentence labels.